Oil Spill Detection in PolSAR Imagery Using Composite Scattering Power Entropy and Multiscale Hybrid Feature Fusion Network

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Abstract

This paper proposes a polarimetric SAR (PolSAR) oil spill detection framework that integrates a novel composite polarimetric scattering power entropy feature with a multiscale hybrid feature fusion network (MSHFFNet). The method addresses challenges caused by oil spill look-alikes, elongated slick structures, and small-scale oil films. Experimental results using Radarsat-2 data demonstrate superior detection accuracy and boundary delineation compared to existing methods.

1 Introduction

Marine oil spills pose severe threats to marine ecosystems and coastal economies. Synthetic aperture radar (SAR) is widely used for oil spill monitoring due to its all-weather, day-and-night imaging capability. In SAR imagery, oil slicks appear as dark regions because oil suppresses capillary and short gravity waves.

However, many natural phenomena such as low wind areas, plant oil, internal waves, and ship wakes also produce dark regions, known as look-alikes. These similarities significantly limit the effectiveness of single-polarization SAR for oil spill detection.

Polarimetric SAR (PolSAR) provides richer scattering information that helps distinguish oil slicks from look-alikes. Despite this advantage, existing polarimetric features and detection networks still struggle with small-scale and elongated oil spills. This paper addresses these issues through feature innovation and network design.

2 Related Work

Early PolSAR-based oil spill detection relied on individual polarimetric features such as entropy, anisotropy, span, and Bragg scattering ratios. Although effective in simple scenarios, these features often fail in complex environments due to overlapping scattering characteristics.

Traditional machine learning methods combine multiple polarimetric features but depend heavily on handcrafted features and lack deep semantic understanding. Deep learning approaches improve representation capability but often struggle with elongated oil slicks, boundary accuracy, and interference from look-alikes.

This paper proposes a unified solution combining discriminative polarimetric features with a deep multiscale segmentation network.

3 Composite Polarimetric Scattering Power Entropy

Oil slicks alter the scattering mechanism of the sea surface, increasing scattering randomness and complexity. To capture this behavior, the paper introduces composite polarimetric scattering power entropy, denoted as H_c.

The feature is constructed by multiplying:

- Polarimetric entropy H, derived from Cloude decomposition, which measures scattering randomness.
- Scattering power entropy H_p, derived from Freeman decomposition, which measures scattering mechanism complexity.

$$H_c = H \times H_p$$

This composite feature enhances the contrast between oil slicks, seawater, and look-alikes. Quantitative evaluations using Jeffreys–Matusita distance and probability density functions confirm that H_c provides superior class separability compared to traditional polarimetric features.

4 Dataset Description

The experiments use fully polarimetric Radarsat-2 SAR data from three scenarios:

- European North Sea controlled oil spill experiment
- Gulf of Mexico oil spill image I
- Gulf of Mexico oil spill image II

The datasets include crude oil, emulsified oil, plant oil, and seawater. Expert-labeled ground truth data are used for training and evaluation. Lee filtering is applied to reduce speckle noise prior to feature extraction.

5 Multiscale Hybrid Feature Fusion Network

The proposed MSHFFNet follows an encoder–decoder architecture. The encoder uses ResNet34 to extract multilevel features from composite entropy H_c and polarimetric intensity channels T_{11}, T_{22}, and T_{33}.

The decoder integrates two key modules:

- Hybrid pooling module
- Feature fusion module

This design allows the network to capture both fine spatial details and long-range contextual information.

6 Hybrid Pooling Module

Standard square pooling is effective for compact objects but performs poorly for elongated oil slicks. To address this limitation, the hybrid pooling module combines:

- Square pooling for local contextual information
- Strip pooling for long-range dependency modeling

Strip pooling aggregates information along horizontal or vertical directions, improving detection of long and narrow oil slicks while reducing interference from surrounding seawater.

7 Feature Fusion Module

Small-scale oil slicks require accurate boundary and texture preservation. The feature fusion module combines shallow and deep features using spatial squeeze-and-excitation (sSE) blocks.

Shallow features preserve edge and texture information, while deep features capture semantic context. The sSE blocks assign spatial importance weights, suppressing background noise and highlighting oil spill regions.

This fusion strategy significantly improves detection accuracy for thin and fragmented oil slicks.

8 Loss Function and Evaluation Metrics

To address class imbalance between oil spills and seawater, a combined loss function is used:

- Dice loss for region overlap optimization
- Focal loss for hard-sample emphasis

Performance is evaluated using Overall Accuracy (OA), Average Accuracy (AA), F1 score, and mean Intersection over Union (mIoU).

9 Methodology

The complete detection methodology consists of the following steps:

1. Apply speckle filtering to PolSAR images.
2. Extract polarimetric entropy H, scattering power entropy H_p, and compute H_c.
3. Combine H_c with polarimetric intensity channels.

4. Extract multiscale features using ResNet34 encoder.
5. Capture elongated oil slick structures using hybrid pooling.
6. Fuse shallow and deep features using sSE-based fusion.
7. Train the network using Dice and Focal loss.
8. Generate pixel-level oil spill segmentation maps.

This end-to-end framework jointly exploits physical scattering properties and deep semantic features.

10 Performance Analysis

Extensive experiments demonstrate that the proposed method outperforms RF, Improved-2D-CNN, Dual-EndNet, and DRSNet across all datasets.

Key results include:

- Higher OA, AA, F1, and mIoU scores
- Superior boundary accuracy for elongated oil slicks
- Reduced false alarms caused by look-alikes
- Improved detection of small-scale oil spill regions

Ablation studies confirm that both the hybrid pooling module and feature fusion module are essential for optimal performance. Additional experiments validate the robustness of H_c under noise and varying signal-to-noise ratios.

11 Discussion

The composite entropy feature effectively suppresses interference from plant oil and seawater. The multiscale hybrid network design addresses limitations of conventional CNNs in handling elongated and fragmented structures.

Noise analysis shows that copolarized channels provide more reliable information than cross-polarized channels, supporting the choice of intensity inputs used in the network.

12 Conclusion

This paper presents a robust PolSAR oil spill detection framework combining composite polarimetric scattering power entropy with a multiscale hybrid feature fusion network. The proposed method significantly improves discrimination between oil slicks and look-alikes while enhancing boundary accuracy.

Experimental results on Radarsat-2 data confirm state-of-the-art performance. The framework is well suited for large-scale, operational marine oil spill monitoring and can be extended to other PolSAR-based environmental applications.